AI for medical imaging
Quizz
2024-10-01

Firstname SURNAME: Student number:

Answer all questions succinctly, to the best of your ability. The quality of the prose will not be graded; but its clarity and accuracy, very much.

1 – One hospital has 1M heart scans of the short axis view of US, and hospitals 2 and 3 have 1K scans for the same. How can we develop a model which can help all the hospitals without sharing the data among them? (2pt)

Federated learning

2 – You have a dataset with images of various resolution (i.e., $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N}, \exists n \neq m \in \mathcal{D} : \Omega_n \neq \Omega_m$). Assuming you are using a batch size of 1 and no sub-patching is needed (your GPU has enough memory), do you *need* to resize all samples to be able to use a Fully Convolutional Network? Motivate your answer. (2pt)

No need to resize (FCN output matches its input in size)

3 – For a given sample, the label of class k is empty ($\Omega_y^{(k)} = \emptyset$), and so is its corresponding prediction ($\Omega_{s_\theta}^{(k)} = \emptyset$). Compute the corresponding Dice score (giving details) and explain how you propose to modify the definition and your implementation to solve the difficulty. Wow could it bias your metric? (2pt)

half point if this one missing

$DSC(s_\theta, y) = \frac{0}{0+0} = \frac{0}{0} = \text{infinity}$, $F_{i0} \Rightarrow \frac{0+\varepsilon}{0+0+\varepsilon} = 1$, ε very small.

If/else also acceptable.

4 – Imagine you have a model trained to detect abnormalities in brain MRI scans, but now you need it to work on detecting issues in lung CT scans. The model struggles because the two types of images are very different. How can you make the model work well with lung CT scans without retraining it from scratch? (2pt)

Domain adaptation

5 – Remembering the partial derivatives of the Cross-entropy loss and Dice loss, with respect to specific voxels predictions ($\frac{\partial \mathcal{L}(y, s_\theta)}{\partial s_\theta^{(i,l)}}$). Cite one of the two key differences on how the two losses supervise a neural network. (2pt)

- DSC gradient much smaller ($\left|\frac{\partial \mathcal{L}_{DSC}}{\partial s_{\theta}^{(i,l)}}\right| \ll \left|\frac{\partial \mathcal{L}_{CE}}{\partial s_{\theta}^{(i,l)}}\right|$)

- DSC gradient takes only two different values over the image (per class)

6 – You are scanning a patient in a MRI twice, with one week between each scan. What type of registration (rigid or non-rigid) would you expect to be required depending on the organ you want to compare: kidney, heart, femur, ribs, lungs. Motivate your choices. (2pt)

rigid
- femur

either accepted
- ribs

non-rigid
- kidney
- heart
- lungs

7 – How do you get training labels for learned registration (e.g., Voxelmorph)? Explain. (3pt + 1 bonus)

Trick question: no label as it uses unsupervised functions (image similarity + regularizer).

Bonus point: mention annotated landmarks or segmentation

8 – You trained a deep network on data collected from the clinical practice in your hospital, and want to deploy it in other hospitals (regulatory approval notwithstanding). Between a model using ultrasound images, or CT scans, which one you expect to have the most troubles to generalize in those new sites. Explain your reasoning. (1pt)

Ultrasounds as it is manually operated

9 – You are developing a novel, ground-breaking neural network architecture, and per the grants funding your research it has to be "open" and "reproducible". Your code must be made public, and the paper should be open-access. Yet, some of the datasets (not all) you used for validation are private from your hospital and cannot be made public. Would you consider your research reproducible? Explain your reasoning. (3pt)

graded quite generously

Full point when reminding that "reproducible" is ill defined, and that some results can be reproduced, & the method can still be applied to other tasks.

10 – Give at least three statistics used for fingerprinting the data in the nnU-Net pipeline. (1pt)

See paper.
- Spacings
- Modality
- Shape
- intensity distribution

11 – (Bonus) What is the main reason John VON NEUMANN gave Claude SHANNON to name his information measure "entropy"? (1pt)

"[...] no one really knows what it is, so in a debate you'll always have the advantage"